When a wave propagates it is treated as the propagation of wavefront. The wavefront is always perpendicular to the direction of the propagation of the wave. As the direction of ray is in the direction of propagation of the wave, the wavefront is always perpendicular to the ray as shown in Figure 7.1(b).

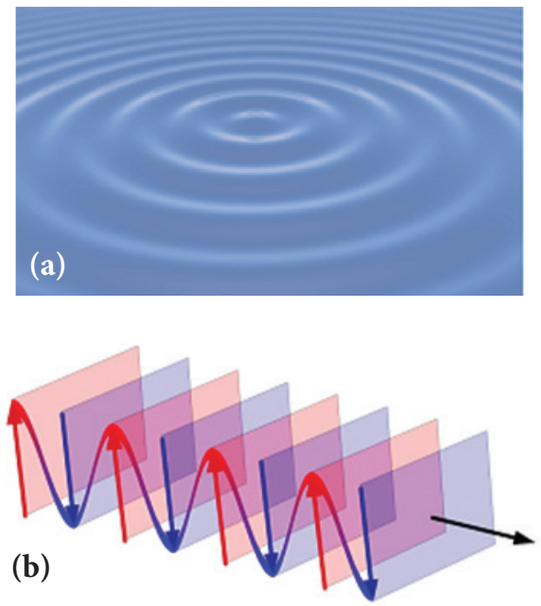


Figure 7.1 (a) Ripples on water surface (b) Wavefront and ray

The shape of a wavefront observed at a point depends on the shape of the source and also the distance at which the source is located. A point source located at a finite distance gives spherical wavefronts. An extended (or) line source at finite distance gives cylindrical wavefronts. Any source that is located at infinity gives plane wavefront as shown in Figure 7.2.

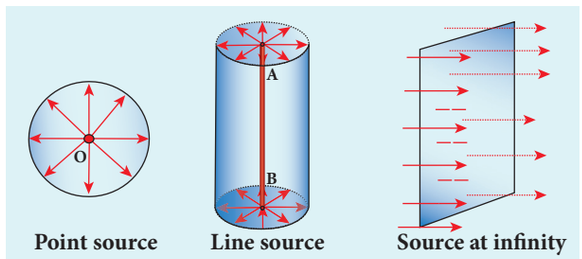


Figure 7.2 Wavefronts

7.2.2 Huygens' Principle

Huygens principle is basically a geometrical construction which gives the shape of the wavefront at any time if we know its shape at $t = 0$. **According to Huygens principle, each point on the wavefront behaves as the source of secondary wavelets spreading out in all directions with the speed of the wave. These are called as secondary wavelets. The envelope to all these wavelets gives the position and shape of the new wavefront at a later time**. Thus, Huygens' principle explains the propagation of a wavefront.

The propagation of a spherical and plane wavefront can be explained using Huygens' principle. Let, AB be the wavefront at a time, $t = 0$. According to Huygens' principle, every point on AB acts as a source of secondary wavelet which travels with the speed of the wave (speed of light c). To find the position of the wavefront after a time t, circles of radius equal to ct are drawn with points P, Q, R ... etc., as centers on AB. The forward envelope (or) the tangent $A'B'$ of the small circles is the new wavefront at that instant t. The wavefront $A'B'$ will be a spherical wavefront from a point object which is at a finite distance as shown in Figure 7.3(a) and it is a plane wavefront if the source of light is at a large distance (infinity) as shown in Figure 7.3(b).

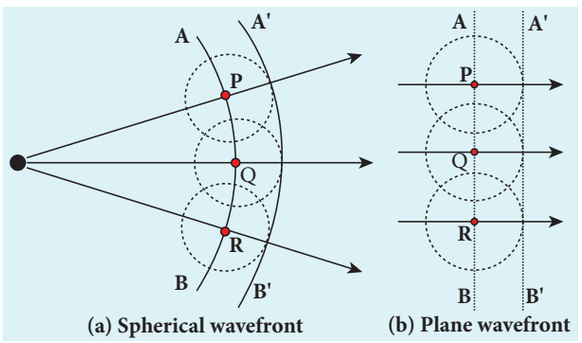


Figure 7.3 Huygens' Principle

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